

DXF File Format Document

1. Definition

DXF (Drawing Exchange Format) is a CAD file format developed by Autodesk to enable data exchange between different CAD software.

2. Basic Information

- Full Name: Drawing Exchange Format
- Extension: .dxf
- Category: CAD File
- Developer: Autodesk
- File Type: ASCII or Binary

3. Purpose

DXF files are used to share design data, store 2D/3D drawings, and ensure compatibility across CAD programs.

4. File Structure

- HEADER: General information
- TABLES: Layers and styles
- BLOCKS: Reusable components
- ENTITIES: Drawing objects
- OBJECTS: Additional data

5. Types

ASCII DXF: Readable, larger size

Binary DXF: Compact, faster, not readable

6. Uses

Engineering, architecture, CNC, laser cutting

7. Software

AutoCAD, SolidWorks, Fusion 360, LibreCAD

8. Advantages

Widely supported, flexible

9. Limitations

Large size, limited advanced feature support

10. Summary

DXF is a standard format for CAD data exchange.